

Technology Stack

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack

S.No	Technology Category	Technology Used	Purpose / Justification
1	Programming Language	Python	Develops ML model, backend logic and preprocessing.
2	Frontend	HTML5, CSS3, Bootstrap 5	Responsive interface for weather input and prediction results.
3	Backend Framework	Flask	Handles requests and connects the UI with the ML model.

4	Machine Learning	Random Forest Classifier	Best-performing model for flood prediction.
5	Data Processing	Pandas, NumPy	Data cleaning, preprocessing and numerical analysis.
6	Data Visualization	Matplotlib, Seaborn	EDA using charts, box plots and heatmaps.
7	Feature Scaling	StandardScaler	Standardizes features before prediction.
8	Model Persistence	Joblib	Stores and loads the trained model and scaler.
9	Dataset Storage	CSV Dataset	Stores historical weather and rainfall data.
10	Development Environment	VS Code, Jupyter Notebook	Development, experimentation and debugging.
11	Version Control	Git & GitHub	Source code management and collaboration.

12	Deployment	IBM Cloud (Future Scope)	Future cloud deployment of the application.
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